

8.05 Surfaces and Partial Differentiation

Surfaces in 3-D are explored, including contours and sections, and applying partial differentiation to find stationary points, and to classify them using the Hessian matrix.

3. visualisation. Learners could use appropriate software to visualise situations in 3-D relating to surfaces, and to linear transformations and invariance.
4. Spreadsheets: Learners could use spreadsheet software to investigate sequences and series, for modelling and to generate tables of values for functions.

Content of Additional Pure Mathematics (Optional paper Y545)

When this course is being co-taught with AS Level Further Mathematics A (H235) the 'Stage 1' column indicates the common content between the two specifications and the 'Stage 2' column indicates content which is particular to this specification. In each section the OCR reference lists 'Stage 1' statements before 'Stage 2' statements.

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...
8.01 Sequences and Series			
8.01a	Recurrence relations	a) Be able to work with general sequences given either as recurrence relations or by position-to-term (closed form) formulae $u_n = f(n)$. <i>The notation $\{u_n\}$ for sequences, which may or may not include a zeroth term, should be recognised.</i>	
8.01b		b) Use induction to prove results relating to both sequences and series.	
8.01c	Properties of sequences	c) Understand and be able to describe various possibilities for the behaviour of sequences. <i>Learners are expected to be able to use the terms "periodic", "convergence", "divergence", "oscillating", "monotonic".</i> <i>Note that a periodic sequence with period two may be referred to as "oscillating", but that both convergent and divergent sequences can oscillate. "Divergence" can refer to sequences that are bounded or unbounded.</i>	
8.01d		d) Identify and be able to use the limit of the n th term of a sequence as $n \rightarrow \infty$, including steady-states. <i>Includes forming sequences from other sequences, for example, finding differences or ratios of successive terms of a sequence.</i> <i>[Rates of convergence are excluded.]</i>	

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8.01e	Fibonacci and related numbers	e) Be able to work with the Fibonacci numbers (and other Fibonacci-like sequences, such as the Lucas numbers), and understand their properties. <i>Includes recognising and using the properties of ϕ, both numerical and algebraic, and its role in the Fibonacci sequence.</i>	
8.01f 8.01g	Solving recurrence systems	f) Be able to solve a first-order linear recurrence relation with constant coefficients, using the associated auxiliary equation and complementary function. <i>Includes finding both general and particular solutions.</i> <i>Includes homogeneous and non-homogeneous recurrence relations of the form $u_{n+1} = au_n + f(n)$, where $f(n)$ may be a polynomial function or of the form dk^n.</i> <i>Includes knowing the terms, “closed form” and “position-to-term”.</i> <i>Includes understanding that a “recurrence system” consists of a “recurrence relation”, an “initial condition” and the range of the variable n.</i>	g) Be able to solve a second-order linear recurrence relation with constant coefficients, using the associated auxiliary equation and complementary function. <i>Includes finding both general and particular solutions.</i> <i>Includes the cases when the roots of the auxiliary equation are:</i> (i) <i>distinct and real,</i> (ii) <i>repeated,</i> (iii) <i>complex.</i> <i>Includes homogeneous and non-homogeneous recurrence relations of the form $u_{n+2} = au_{n+1} + bu_n + f(n)$, where $f(n)$ may be a polynomial function or of the form dk^n.</i> <i>Learners should be aware that this topic is the discrete analogue of the work on differential equations in H245 section 4.10.</i>
8.01h 8.01i	Modelling	h) Be able to apply their knowledge of recurrence relations to modelling. <i>Includes birth- and/or death-rates and the use of the $\text{INT}(x)$ function for discrete models. Learners may find it useful to have a calculator with this function, but large numbers of repeated applications will not be required in the assessment.</i>	i) Be able to extend their knowledge of modelling to second order recurrence relations.

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8.02 Number Theory			
8.02a	Number bases	a) Understand and be able to work with numbers written in base n, where n is a positive integer. <i>The standard notation for number bases will be used.</i> <i>i.e. 2013_n will denote the number $2n^3 + n + 3$ (with $n > 3$ in this example) and the letters $A-F$ will be used to represent the integers 10-15 respectively when $11 \leq n \leq 16$.</i>	
8.02b	Divisibility tests	b) Be able to use (without proof) the standard tests for divisibility by 2, 3, 4, 5, 8, 9 and 11. <i>Includes knowing that repeated tests can be used to establish divisibility by composite numbers.</i>	
8.02c		c) Be able to establish suitable (algorithmic) tests for divisibility by other primes less than 50. <i>For integers a and b, the notation $a \mid b$ will be used for “a divides exactly into b” (“a is a factor of b”, “b is a multiple of a”, etc.).</i>	
8.02d	The division algorithm	d) Appreciate that, for any pair of positive integers a, b with $0 < b \leq a$, we can uniquely express a as $a = bq + r$ where q (the quotient) and r (the residue, or remainder, when a is divided by b) are both positive integers and $r < b$.	

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8.02e 8.02g 8.02f 8.02h	Finite (modular) arithmetics	e) Understand and be able to use finite arithmetics (the arithmetic of integers modulo n for $n \geq 2$). f) Be able to solve single linear congruences of the form $ax \equiv b \pmod{n}$.	g) Be able to calculate quadratic residues and solve, or prove insoluble, equations involving them. h) Be able to solve simultaneous linear congruences of the form $ax \equiv b \pmod{n}$. <i>No more than three simultaneous linear congruences will be used. Use of the Chinese remainder theorem will be allowed but not required.</i>
8.02i 8.02j	Prime numbers	i) Understand the concepts of prime numbers, composite numbers, highest common factors (hcf), and coprimality (relative primeness). <i>Knowledge of the fundamental theorem of arithmetic will be expected, but proof of the result will not be required.</i> j) Know and be able to apply the result that $a \mid b$ and $a \mid c \Rightarrow a \mid (bx + cy)$ for any integers x and y. <i>Includes using this result, for example to test for common factors or coprimality.</i>	
8.02k	Euclid's lemma	k) Know and be able to use Euclid's lemma: if $a \mid rs$ and $\text{hcf}(a, r) = 1$ then $a \mid s$.	

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8.02l	Fermat's little theorem		l) Know and be able to use Fermat's little theorem in the following forms: - if p is prime and $\text{hcf}(a, p) = 1$, then $a^{p-1} \equiv 1 \pmod{p}$. - if p is prime $a^p \equiv a \pmod{p}$. <i>Includes recognising that if p is prime then Fermat's theorem holds, but that the converse is not true and being aware of the existence of "pseudo-primes" to base a.</i> [The proof is excluded.] [Carmichael numbers are excluded.]
8.02m 8.02n	The order of a modulo p		m) Know and be able to use the fact that $p - 1$ is necessarily the least positive integer n for which $a^n \equiv 1 \pmod{p}$. n) Know that the n with this property, called the order of a modulo p, is a factor of $p - 1$ and be able to use this in specific cases.
8.02o	Binomial theorem		o) Be able to use the binomial theorem to show that $(a + b)^p \equiv a^p + b^p \pmod{p}$ for prime p , and state the result.
8.03 Groups			
8.03a 8.03b	Binary operations	a) Be able to work with binary operations and their properties when defined on given sets. <i>Includes knowing and understanding the terms "associativity" and "commutativity".</i> b) Be able to construct Cayley tables for given finite sets under the action of a given binary operation. <i>Multiplicative notation and/or terminology will generally be used, when appropriate.</i>	

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8.03c	Definition of a group	c) Recall and be able to use the definition of a group, for example to show that a given structure is, or is not, a group. <i>e.g. Questions may be set on groups of integers modulo n (for $n \geq 2$), functions, matrices, transformations, the symmetries of given geometrical shapes and complex numbers.</i> <i>Groups may be referred to in either of the forms:</i> 1. <i>by the given set and associated binary operation $(G, \circ)$,</i> 2. <i>as “G”, where the operation is understood, or</i> 3. <i>as “the set G with the operation $\circ$”.</i> <i>To include knowing the meaning of the terms “identity” and “closed”, and that in an abelian group the operation is commutative.</i>	
8.03d		d) Recognise and be able to use the Latin square property for group tables.	
8.03e	Orders of elements and groups	e) Recall the meaning of the term “order”, as applied both to groups and to elements of a group, and be able to determine the orders of elements in a given group. <i>Includes knowing and being able to use the fact that the order of an element is a factor of the order of the group.</i>	
8.03f	Subgroups	f) Understand and be able to use the definition of a subgroup, find subgroups and show that given subsets are, or are not, proper subgroups.	
8.03g	Cyclic groups	g) Recall the meaning of the term “cyclic” as applied to groups.	

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8.03h	Generators	h) Understand that a cyclic group is generated by “powers” of a single element (generator), that there may be more than one such element within a group, and that other (non-cyclic) groups may be generated by two or more elements along with their “powers” and “products”.	
8.03i 8.03j	Properties of groups	i) Be familiar with the structure of finite groups up to, and including, order seven, and be able to apply this knowledge in solving problems.	j) Be able to work with groups of higher finite order, or of infinite order. <i>No particular prior knowledge of specific groups, or their structures, will be expected.</i>
8.03k	Lagrange’s theorem		k) Recall and be able to apply Lagrange’s theorem concerning the order of a subgroup of a finite group. <i>The proof of the theorem is not required, but a clear statement of the result may be expected.</i>
8.03l	Isomorphism		l) Be able to determine whether two given groups are, or are not, isomorphic using informal methods. <i>e.g. By noting disparities between the orders of elements.</i>
8.03m	Abstract groups		m) Be able to work with groups defined by their algebraic properties. <i>Includes using algebraic methods to establish properties in abstract groups.</i> <i>e.g. To show that any group in which every element is self-inverse is abelian, to establish an identity or complete a Cayley table.</i>

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8.04 Further Vectors			
8.04a	Vector product	a) Understand and be able to use the definition, in geometrical terms, of the vector product and be able to form the vector product in magnitude and direction, and in component form. <i>Includes use of the formula $\mathbf{a} \times \mathbf{b} = \mathbf{a} \mathbf{b} \sin \theta \hat{\mathbf{n}}$, where $\mathbf{a}$, $\mathbf{b}$, $\hat{\mathbf{n}}$, in that order (and the vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$, in that order) form a right-handed triple.</i>	e) Understand and be able to use the definition of scalar triple product, and be able to use it to calculate volumes of tetrahedra and parallelepipeds. <i>Includes the notation $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$. Includes understanding the significance of $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) =$</i>
8.04b		b) Understand the anti-commutative and distributive properties of the vector product.	
8.04c		c) Be able to use the vector product to calculate areas of triangles and parallelograms.	
8.04e		d) Understand the significance of $\mathbf{a} \times \mathbf{b} = \mathbf{0}$. <i>e.g. The equation of a line in the form $(\mathbf{r} - \mathbf{a}) \times \mathbf{d} = \mathbf{0}$.</i>	
8.04d			
8.05 Surfaces and Partial Differentiation			
8.05a 8.05b	3-D surfaces	a) Be able to work with functions of two variables, given either explicitly in the form $z = f(x, y)$ or implicitly in the form $g(x, y, z) = c$, and understand and use the fact that this equation, and its partial derivatives, relate to a 3-D surface. <i>An informal understanding only of how the partial derivatives relate to the surface is required.</i> <i>Functions $f(x, y)$ will involve sums and products of powers of x and y only. Issues relating to domains and ranges will not be considered beyond the appreciation that, for example the surface $z = \frac{x}{y}$ has no point at which $y = 0$.</i>	b) Extend their knowledge of surfaces to those defined by functions of more than two variables, and incorporate trigonometric functions, logarithms and exponential functions.

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8.05c	Sections and contours	c) Be able to sketch sections and contours, and know how these are related to the surface. <i>i.e. Sections of the form $z = f(a, y)$ or $z = f(x, b)$ and contours of the form $c = f(x, y)$.</i>	
8.05d	Partial differentiation	d) Be able to find first and second derivatives, including mixed derivatives. <i>Learners will be expected to recognise and use both notations for first- and second-order partial derivatives, including mixed ones.</i> <i>e.g. $\frac{\partial f}{\partial x}$, $\frac{\partial^2 f}{\partial x \partial y}$ and f_x, f_{xy}.</i> <i>Includes the Mixed derivative theorem; namely, that $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$ or $f_{xy} = f_{yx}$ for suitably well-defined, continuous functions f.</i>	
8.05e 8.05f	Stationary points	e) Understand and be able to apply the concept that stationary points of z arise when $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$ (or $f_x = f_y = 0$) and that these can be maxima, minima or saddle-points. <i>Learners should know and understand the basic properties of these stationary points.</i> <i>Learners will only be required to find stationary points, but will not be required to determine their natures.</i>	f) Be able to determine, for 3-D surfaces given in the form $z = f(x, y)$, the nature of maxima, minima and saddle-points by means of the sign of the determinant of the matrix of second partial derivatives (the Hessian Matrix), $\mathbf{H} = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix};$ namely, that: 1. if $\mathbf{H} > 0$ and $f_{xx} > 0$, there is a (local) minimum; 2. if $\mathbf{H} > 0$ and $f_{xx} < 0$, there is a (local) maximum; 3. if $\mathbf{H} < 0$ there is a saddle-point;

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8.05g	Tangent planes		g) Be able to determine, for 3-D surfaces given in the form $z = f(x, y)$, the equation of a tangent plane to the surface at a given point $(x, y, z) = (a, b, f(a, b))$ using the formula $z = f(a, b) + (x - a)f'_x(a, b) + (y - b)f'_y(a, b)$